

SMART VAN SALES MANAGEMENT SYSTEM

Smart Van Sales Management System

End-to-end direct-store-delivery platform for food & beverage

Hot Sale

FEFO Stock

Route Optimisation

Local-first AI

From the factory to the shop shelf — one system

v1.0.0

THE PROBLEM

At day end, everyone quotes a different number.

When van stock, invoices raised, cash collected and the physical count are each kept separately, nobody can trace where the evening's discrepancy came from.

Four places it breaks

Where van sales operations most reliably lose money

Guessed van stock

What was loaded is in a notebook, what was sold is on receipts. When they disagree at night, the variance belongs to nobody.

Expiry leakage

Short-dated stock sits on the shelf while a newer batch is sold. The write-off is discovered only at disposal.

Route blindness

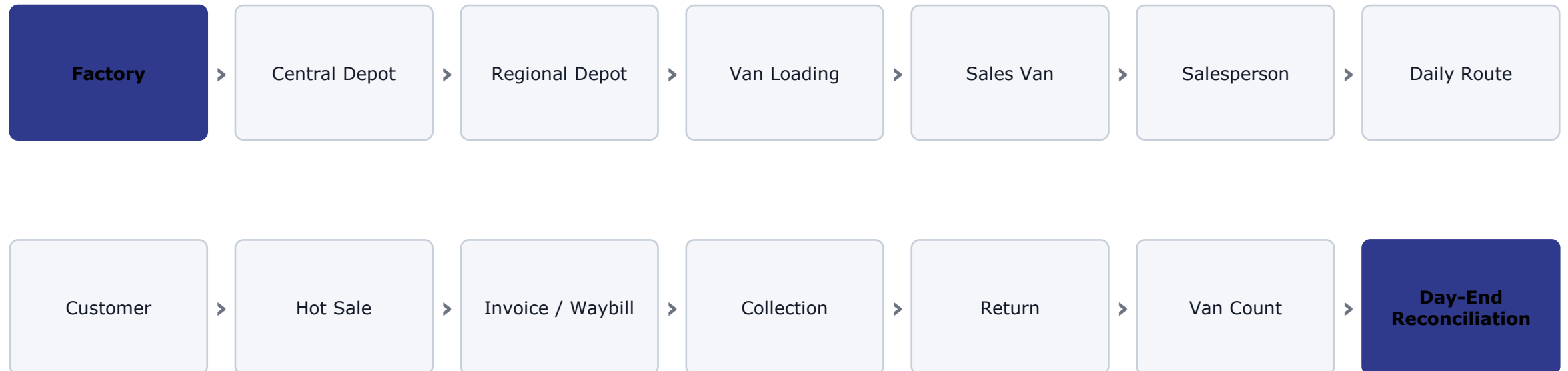
Routes are built from habit. Nobody measures the relationship between kilometres driven and revenue earned.

Credit risk

The credit limit lives on paper; the office learns about the over-limit sale after the rep has already made it.

The chain this system covers

Every link feeds from the previous one — no manual re-entry anywhere



The measure of the system

Figures measured from the running system, not targets

118

database tables

293

API endpoints

49

interface screens

207

permissions

37

business services

21

built-in reports

942

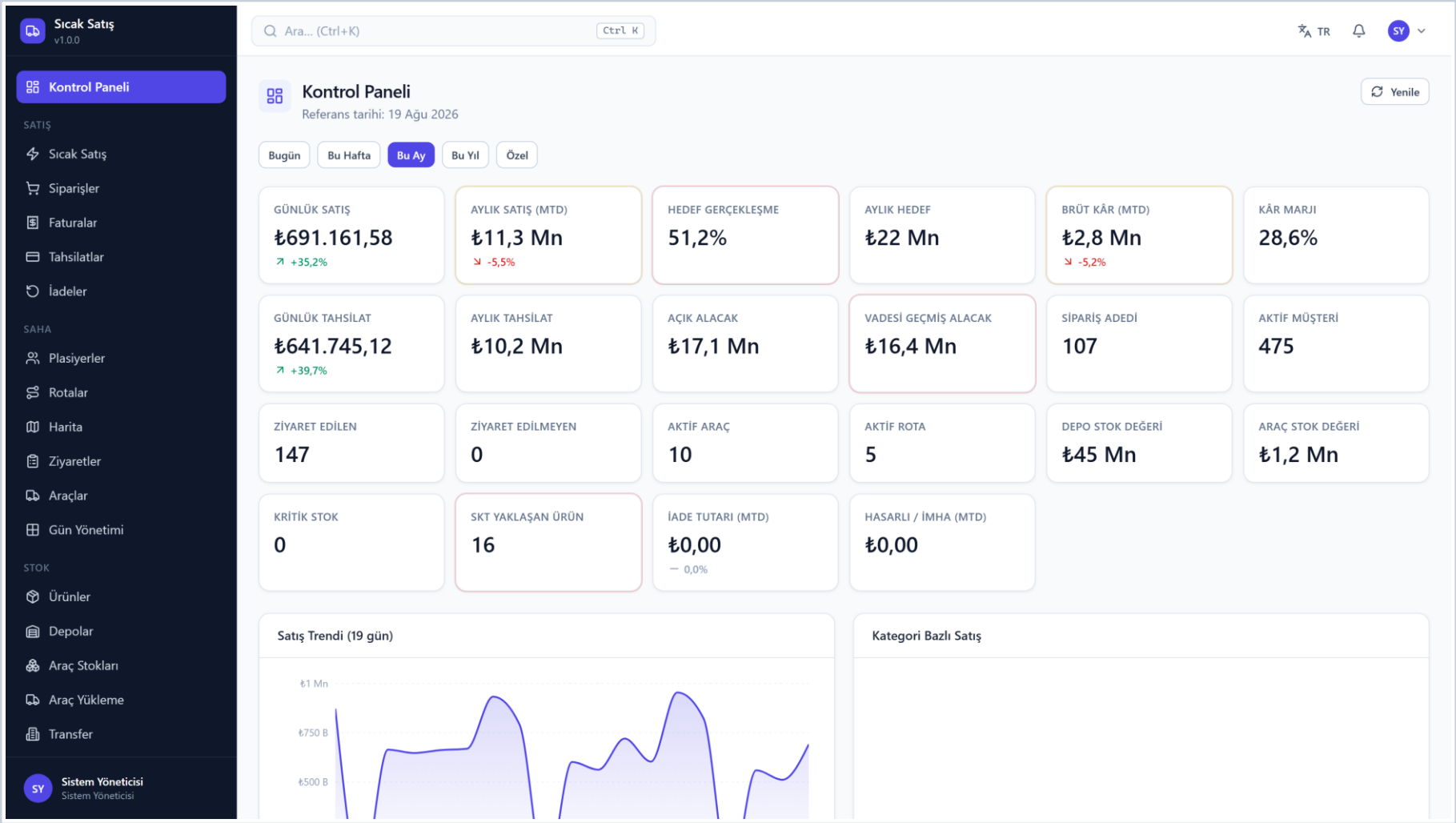
database indexes

2

languages (TR / EN)

The dashboard

20 KPIs and 7 live charts — panel and reports share one aggregation



01 **Hot Sale**

The screen a rep opens hundreds of times a day

Six taps at the customer's door

Built to be used one-handed, standing in a shop

1 · Pick the customer

Balance, credit limit, risk and recent purchases all on the same screen.

2 · See the van stock

Only what is physically on the van; short-dated items carry a coloured warning badge.

3 · Read the AI suggestion

From average consumption and days since the last purchase, the system estimates what has run out.

4 · Build the basket

Enter quantity and unit; campaigns apply themselves and free goods appear as their own line.

5 · Take payment

Cash, card, transfer, cheque or open account. A cheque does not reduce the balance until it clears.

6 · Complete the sale

Order, delivery, invoice, ledger entry and stock issue are created in one go.

HOT SALE

The salesperson's screen

- Picking a customer shows balance, credit limit, risk score and past orders
- The middle column is the van's real stock; short-dated items carry a badge
- Campaigns apply themselves — free goods appear as their own line
- Totals compute live: gross, line discount, campaign, VAT
- Payment method is chosen and the sale posts in one transaction

Sıcak Satış
v1.0.0

Kontrol Paneli

SATIŞ

Sıcak Satış

Siparişler

Faturalar

Tahsilatlar

İadeler

SAHA

Rotalar

Harita

Ziyaretler

Gün Yönetimi

STOK

Ürünler

Araç Stokları

Araç Yükleme

Sayım

Lot / SKT

CRM

Müşteriler

Hatice Çelik
Plasiyer

Q Ara... (Ctrl+K) Ctrl K

Sıcak Satış

Müşteri Seç

Anadolu Büfe
M00101

Değiştir

Bakiye
Kredi Limiti
Geciken
Risk Skoru
Son Sipariş
Durum

0,00
₺5.800,00
0,00
0
17 Agu 2026
AKTİF

GEÇMİŞ ALIŞVERİŞLER

17.08.2026 · SAT-2026-030099
15.08.2026 · SAT-2026-029998
10.08.2026 · SAT-2026-029510
01.08.2026 · SAT-2026-028804
27.07.2026 · SAT-2026-028346

₺8.226,58
₺13.678,96
₺3.800,16
₺10.459,69
₺2.401,71

AI Önerileri

Taze Vadi Kraker 40 g
Taze Vadi Kraker 40 g: son 14 alımda ortalama 5,7 koli aldı, tipik olarak 7 günde tüketiyor. Son alımdan bu yana 30 gün geçti — stoğunun bitmiş olma olasılığı %99.

+ 144

Günseli Cips 110 g
Günseli Cips 110 g: son 15 alımda ortalama 5,0 koli aldı, tipik olarak 10 günde tüketiyor. Son alımdan bu yana 25 gün geçti — stoğunun bitmiş olma olasılığı %99.

+ 120

Pınaros Soğuk Çay 1000 ml
Pınaros Soğuk Çay 1000 ml: son 12 alımda ortalama 6,2 koli aldı, tipik olarak 13 günde tüketiyor. Son alımdan bu yana 9

+ 1

Araç Stoğu

28

Ürün adı veya SKU

Akdeniz Gıda Cips 65 g
SKU00090
26 koli

Akdeniz Gıda Kek 55 g
SKU00080
91 G
30 koli

Akdeniz Gıda Kola 1000 ml
SKU00013
25 koli

Akdeniz Gıda Kraker 40 g
SKU00006
13 koli

Akdeniz Gıda Süt 200 ml
SKU00043
95 G
11 koli

Akdeniz Gıda Yoğurt 1000 g
SKU00034
23 G
27 koli

Günseli Bisküvi 120 g
SKU00017
21 koli

Günseli Cips 110 g
SKU00023
29 koli

Günseli Kola 500 ml
SKU00079
7 koli

Günseli Kuruyemiş 150 g
24 koli

Sepet

Sepeti boşalt

Akdeniz Gıda Cips 65 g
SKU00090

1

CASE

0

%

Akdeniz Gıda Kek 55 g
SKU00080

1

CASE

0

%

PROMOSYON

Günseli Dondurma 100 ml
4 koli

AKTİF KAMPANYALAR

10 Koli Al 1 Koli Bedava
-₺0,00

Brüt
Satır İskontosu
Kampanya İskontosu
Net
KDV

₺760,08
₺0,00
₺0,00
₺760,08
₺152,02

Toplam

₺912,10

ÖDEME

Nakit

Kredi Kartı

Havale / EFT

Çek

Açık Hesap

Alınan Tutar

DESIGN DECISION

All of it is written, or none of it.

order + delivery + invoice + ledger + payment = 1 transaction

Order → delivery → FEFO stock issue → invoice → ledger → payment is a single database transaction. If any step fails, none of it is written. There is no such thing as a half-finished sale.

Working offline

Coverage is not guaranteed in the field — the system assumes that

Held on the device

With no connection the basket waits on the device and the rep keeps selling.

Safe to resend

Every basket carries a unique id. Submitting the same sale twice never creates a second record.

Reference data cached

Product, customer and campaign lists open without a connection.

Installable app

Installs to a phone or tablet as a PWA — no app store process required.

02 **Stock**

In food distribution, stock means dates

An immutable stock ledger

The structure that settles the day-end argument

Movements are never deleted

Stock movements are only appended; a correction is itself a new movement. History cannot be rewritten.

Balance derives from the ledger

A balance is kept for fast reads but can be rebuilt from the ledger at any moment.

FEFO by default

The nearest-expiry batch leaves first. In food that is not a preference, it is a requirement.

Batch traceability

Which batch went to which customer is recorded — a recall produces a list immediately.

VAN LOADING

AI-assisted loading

- Past consumption of the route's customers, weekday and season effects
- Stock already on the van is subtracted
- Volume and weight capacity cannot be exceeded
- Every line carries its own reason and confidence level
- The suggestion is editable — the rep has the last word

Sıcak Satış
v1.0.0

Kontrol Paneli

SATIŞ

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Siparişler

Faturalar

Tahsilatlar

İadeler

SAHA

Plasiyerler

Rotalar

Harita

Ziyaretler

Araçlar

Gün Yönetimi

STOK

Ürünler

Depolar

Araç Stokları

Araç Yükleme

Transfer

SY Sistem Yöneticisi
Sistem Yöneticisi

Ara... (Ctrl+K) Ctrl K

Araç Yükleme

Yapay zeka destekli yükleme önerisi ve yükleme fişi

Plasiyer * Araç * Kaynak Depo * Yükleme Tarihi *

Plasiyer seçin Araç seçin Depo seçin 19.08.2026

☐ Gün içi ikmal [AI önerisi al](#)

Yükleme Önerisi

Seçili satırlar yüklenir

+ Ürün seçin

Henüz satır eklenmedi

Son Yüklemeler

Kayıt bulunamadı

Kapasite Kullanımı

Hacim 0,0 / 0 L

Ağırlık 0,0 / 0 kg

Koli 0

Yükleme Önerisi 0

Yükleme Fişi Oluştur

Notlar

+ Yükleme Fişi Oluştur

ÖZET

19.08.2026

DAY-END RECONCILIATION

The variance is computed, not debated.

theoretical = opening + loaded + reloaded – sold + returned – wastage
variance = theoretical – physical count

Every term comes from the movement ledger; there is no hand-maintained counter anywhere. A non-zero variance flags the session, writes an adjustment movement and notifies the manager.

A van is a warehouse

We did not write a separate 'van stock' subsystem

Same rules

A van is a warehouse of type VEHICLE. FEFO, counting, transfers and valuation all work on it unchanged.

Created automatically

Saving a vehicle creates its warehouse; there is nothing extra to set up.

Capacity enforced

Volume and weight are checked during loading and an overload is refused.

Why this way

A second stock subsystem would be a copy of the same rules that drifts apart over time.

03 The Field

Routes, visits and position

Route optimisation

Capacity, time windows and service time solved together

Two-layer solver

Google OR-Tools when installed; otherwise the system's own Clarke-Wright + 2-opt solver takes over.

The feature never disappears

A missing 100 MB dependency does not disable routing — the fallback solver always works.

Real constraints

Vehicle volume and weight, customer opening hours, service time, working-day limit and priority customers.

Plan vs actual

Completed, skipped and delayed stops, plus the planned-versus-actual kilometre difference.

THE FIELD

Routes and map

- Daily routes are generated from customers' visit days
- Optimisation solves capacity, time windows and service time together
- Which solver ran is stated on screen
- Planned versus actual kilometres are reported

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Sistem Yöneticisi

Q Ara... (Ctrl+K) Ctrl K

TR SY

Rotalar
Planlama, optimizasyon ve saha takibi

Günlük Rotaları Oluştur

Başlangıç 19.08.2026 Bitiş 19.08.2026 Durum Tüm durumlar Plasiyer Tümü Ara Ara

TARİH	KOD	PLASİYER	ARAÇ	DURAKLAR	PLANLANAN KM	DURUM	TAMA
19.08.2026	ROT027-20260819 Emine Yıldız - WED (19.08.2026)	Emine Yıldız	42 DM 1006	0 / 16	0,0	PLANLANDI	
19.08.2026	ROT021-20260819 Hatice Çelik - WED (19.08.2026)	Hatice Çelik	34 DM 1000	0 / 16	0,0	PLANLANDI	
19.08.2026	ROT015-20260819 Hüseyin Yılmaz - WED (19.08.2026)	Hüseyin Yılmaz	16 DM 1004	0 / 16	0,0	PLANLANDI	
19.08.2026	ROT009-20260819 Emine Kılıç - WED (19.08.2026)	Emine Kılıç	01 DM 1008	0 / 16	0,0	PLANLANDI	
19.08.2026	ROT003-20260819 Ali Yılmaz - WED (19.08.2026)	Ali Yılmaz	07 DM 1002	0 / 16	0,0	PLANLANDI	

Detay için listeden bir rota seçin

04 Analytics

A forecast you cannot measure is a forecast you cannot trust

Demand forecasting

FMCG demand is intermittent — classical methods do badly on it

Classify first

Is the series smooth, intermittent or lumpy? The method follows from the class; no single method is imposed.

Then choose

Croston, SBA, TSB, Holt-Winters or a day-of-week seasonal naive — whichever suits this particular series.

Back-test the choice

Validated with MAE / MAPE / RMSE. The method used and its error are shown on screen.

Zeros count

The calendar is zero-filled; those zeros are what define intermittent demand in the first place.

Dashboard and reports

Panel and report share one aggregation — they cannot disagree

20 KPIs

Daily and monthly sales, target achievement, gross margin, collections, open and overdue receivables, stock value, expiring soon.

7 live charts

Sales trend, category split, top products and salespeople, regional distribution, collections versus sales.

21 built-in reports

Sales, salesperson, customer, SKU, brand, region, route, collections, risk, stock, expiry, wastage, returns, campaigns, profitability, targets.

PDF · Excel · CSV

Written as UTF-8 with BOM so Turkish characters open correctly in Excel with no import step.

ANALYTICS

Demand forecasting

- The series is classified first: smooth, intermittent or lumpy
- The method follows from the class — none is imposed
- The choice is validated by back-testing (MAE / MAPE / RMSE)
- Method used, error and confidence interval are shown on screen

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SY Sistem Yöneticisi
Sistem Yöneticisi

Ara... (Ctrl+K)

Ctrl K

TR

SY

Talep Tahmini

Konu

Ürün

Ara

Ara

Müşteri

Ara

Ara

Plasiyer

Şirket geneli

Ufuk (gün) *

14

Periyot

Günlük

☒ Sonucu kaydet

Tahmini Çalıştır

Şirket geneli

Parametreleri seçip tahmini çalıştırın

05 Artificial Intelligence

Local-first, explainable, budgeted

Three providers, one interface

The router picks a model per task and fails over when one is down

LM Studio — local

Runs on your own machine, costs nothing and no data leaves the building. The default first choice.

NVIDIA NIM — cloud

Steps in when the local model is not enough; over 100 models selected by task type.

Claude — cloud

Configurable for long-context and harder reasoning work.

Budget protection

When the monthly limit is reached the paid providers stop while the local model keeps working.

The AI Sales Manager

Plain-language question → read-only query → justified answer

Example questions

«Show the 10 best-selling reps today» · «Show customers with high collection risk» · «Which customers did we lose in the last 90 days?»

It shows its source

The data behind the answer is visible in an expandable panel — "where did that number come from" is answerable.

Read-only

A single SELECT may be executed. No data-modifying command is accepted.

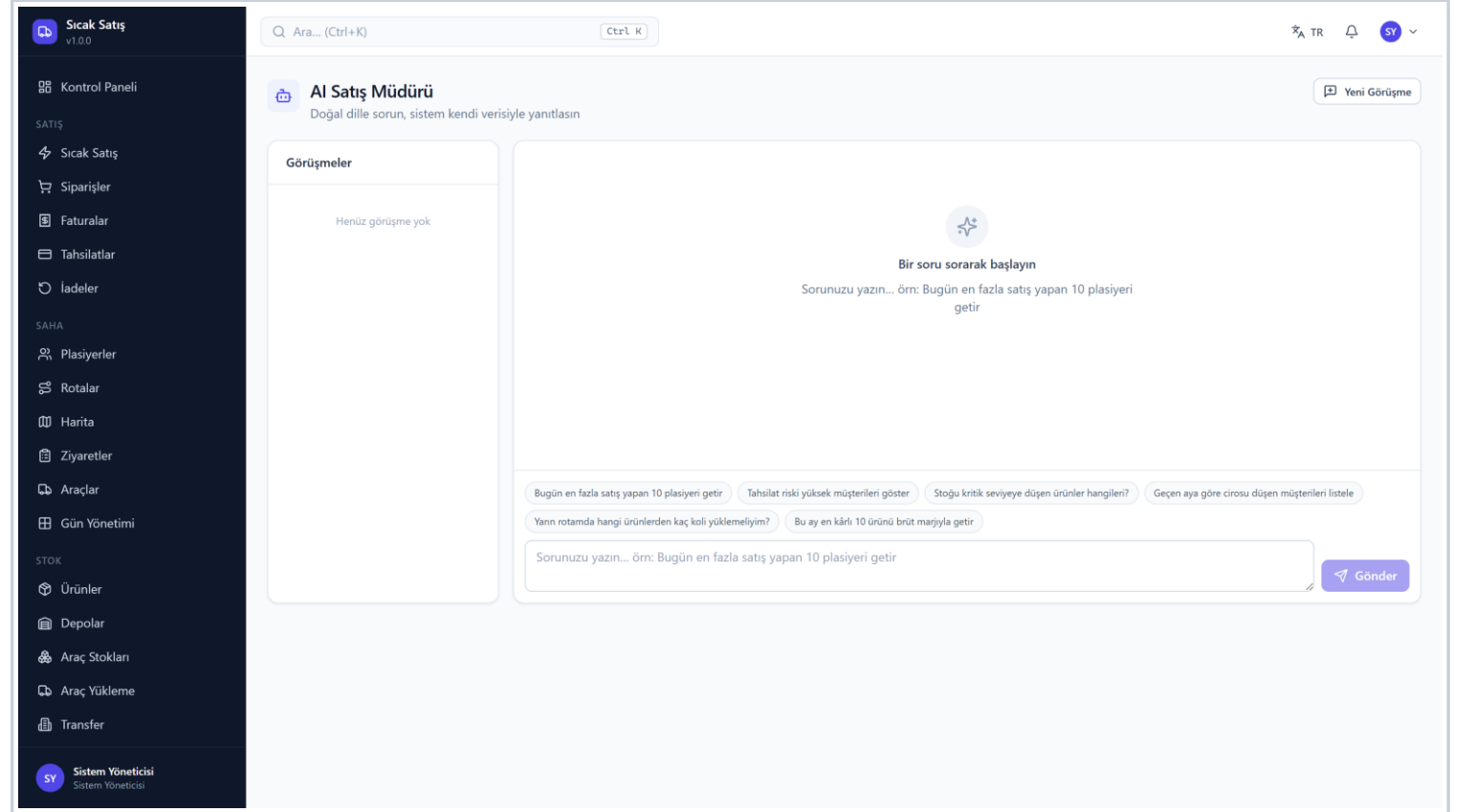
Closed tables

User, session and audit tables are entirely off limits to AI queries.

ARTIFICIAL INTELLIGENCE

The AI Sales Manager

- A question is asked in plain language
- The system produces and runs a read-only query only
- The data behind the answer is visible in an expandable panel
- User, session and audit tables are off limits



Security

Authorisation is checked separately at three layers

19 roles · 207 permissions

Screen, action and data scope are enforced separately. A rep sees only their own records.

No privilege escalation

Nobody can assign a role above their own or grant a permission they do not hold.

Chained audit log

Each entry carries the previous one's digest. Alter a historical row and the chain breaks, reporting where.

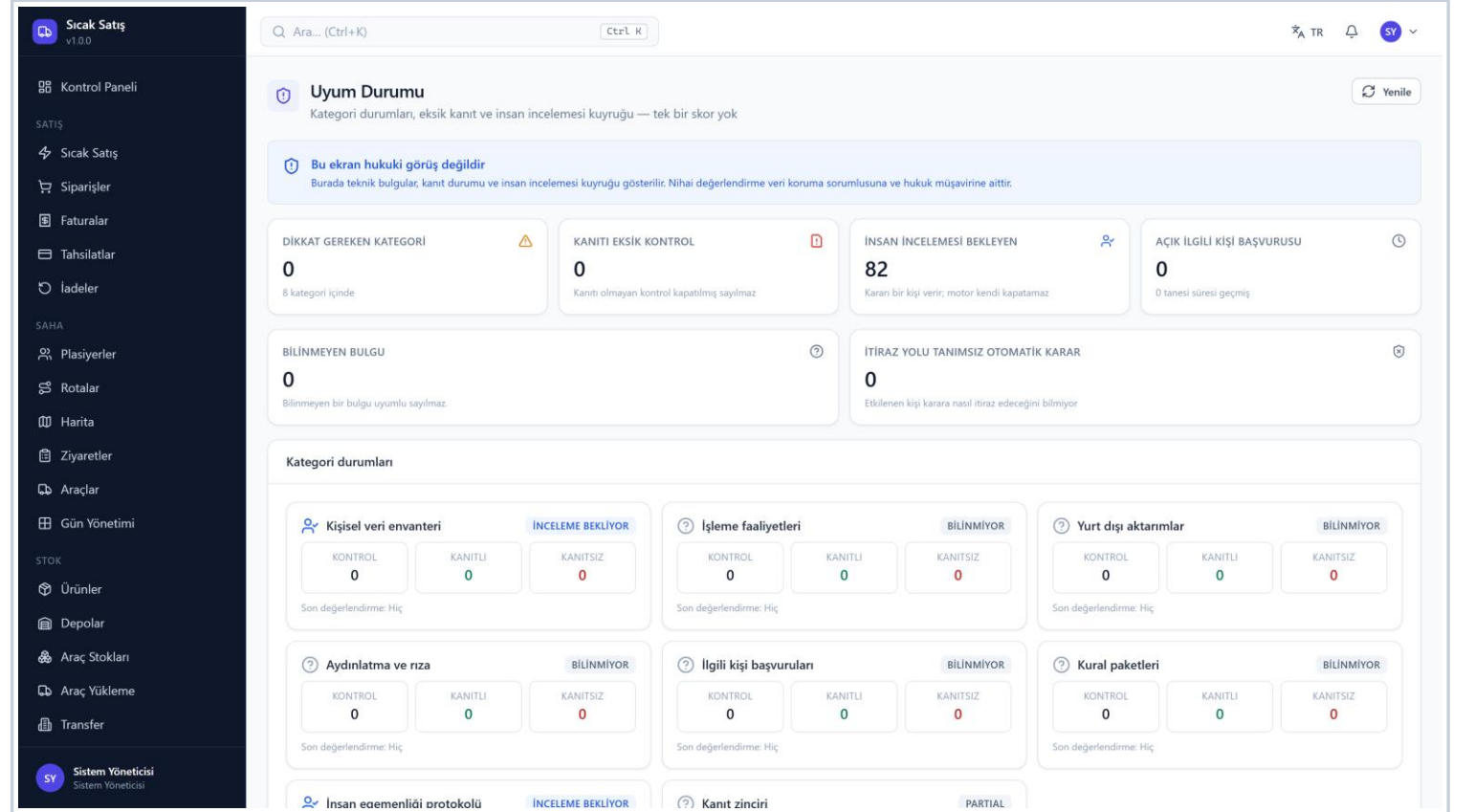
Secrets stay secret

API keys are never stored in the database, are stripped before logging and are masked on screen.

COMPLIANCE

Compliance and human rights layer

- The personal-data inventory is measured from the code, not assumed
- Consent and notice are kept as separate records
- Missing evidence yields 'insufficient evidence', never 'compliant'
- Every decision put to the authority engine — refusals included — leaves a chained receipt and an appeal path



Other screens

The system has 49 screens; six of them here

ID	Ad Soyad	Adres	Telefon	E-posta	Durum	Tarih
M000001	Arasulu Bile	...	...	...	Aktif	...
M000002	Arasulu Bile	...	...	...	Aktif	...
M000003	Arasulu Bile	...	...	...	Aktif	...
M000004	Arasulu Bile	...	...	...	Aktif	...
M000005	Arasulu Bile	...	...	...	Aktif	...
M000006	Arasulu Bile	...	...	...	Aktif	...
M000007	Arasulu Bile	...	...	...	Aktif	...
M000008	Arasulu Bile	...	...	...	Aktif	...
M000009	Arasulu Bile	...	...	...	Aktif	...
M000010	Arasulu Bile	...	...	...	Aktif	...

Customers

ID	Ad	Kategori	Fiyat	Durum
SKU00001	Arasulu Gıda Arasulu 300 ml	...	...	Aktif
SKU00002	Arasulu Gıda Arasulu 150 g	...	...	Aktif
SKU00003	Arasulu Gıda Arasulu 35 g	...	...	Aktif
SKU00004	Arasulu Gıda Arasulu 65 g	...	...	Aktif
SKU00005	Arasulu Gıda Arasulu 250 g	...	...	Aktif
SKU00006	Arasulu Gıda Arasulu 35 g	...	...	Aktif
SKU00007	Arasulu Gıda Arasulu 60 g	...	...	Aktif
SKU00008	Arasulu Gıda Arasulu 100 g	...	...	Aktif
SKU00009	Arasulu Gıda Arasulu 250 ml	...	...	Aktif

Products

Rapor Adı	Tarih	Durum
Arasulu Gıda Arasulu Raporu	...	Aktif
Arasulu Gıda Arasulu Raporu	...	Aktif
Arasulu Gıda Arasulu Raporu	...	Aktif
Arasulu Gıda Arasulu Raporu	...	Aktif
Arasulu Gıda Arasulu Raporu	...	Aktif
Arasulu Gıda Arasulu Raporu	...	Aktif
Arasulu Gıda Arasulu Raporu	...	Aktif
Arasulu Gıda Arasulu Raporu	...	Aktif
Arasulu Gıda Arasulu Raporu	...	Aktif
Arasulu Gıda Arasulu Raporu	...	Aktif

Reports

Tarih	Saat	Durum
19.08.2026	...	Aktif
19.08.2026	...	Aktif
19.08.2026	...	Aktif
19.08.2026	...	Aktif
19.08.2026	...	Aktif
19.08.2026	...	Aktif
19.08.2026	...	Aktif
19.08.2026	...	Aktif
19.08.2026	...	Aktif
19.08.2026	...	Aktif

Day sessions

ID	Ad	Durum
0	Sistem Yöneticisi	Aktif
1	Sistem Sahibi	Aktif
2	Sistem Müdürü	Aktif
3	Sistem Asistanı	Aktif
4	Sistem Destek	Aktif
5	Sistem Yöneticisi	Aktif
6	Sistem Sahibi	Aktif
7	Sistem Müdürü	Aktif
8	Sistem Asistanı	Aktif
9	Sistem Destek	Aktif
10	Sistem Yöneticisi	Aktif

Roles & permissions

Eğitim Adı	Tarih	Durum
Arasulu Gıda Arasulu Eğitim	...	Aktif
Arasulu Gıda Arasulu Eğitim	...	Aktif
Arasulu Gıda Arasulu Eğitim	...	Aktif
Arasulu Gıda Arasulu Eğitim	...	Aktif
Arasulu Gıda Arasulu Eğitim	...	Aktif
Arasulu Gıda Arasulu Eğitim	...	Aktif
Arasulu Gıda Arasulu Eğitim	...	Aktif
Arasulu Gıda Arasulu Eğitim	...	Aktif
Arasulu Gıda Arasulu Eğitim	...	Aktif
Arasulu Gıda Arasulu Eğitim	...	Aktif

Training centre

Technology and setup

Nothing that was not already installed has been made mandatory

Technology

- Python 3.11 · FastAPI · SQLAlchemy 2.0
- React 18 · TypeScript · Vite · Tailwind
- SQLite by default — PostgreSQL optional
- PWA: offline capable, installable
- Redis and Docker are not required

Setup

- setup.ps1 — one-command install
- start.bat — one-click start
- Demo data: 500 customers, 12 months history
- 14-lesson in-app training centre
- TR / EN — instant language switch

PLATFORMS & DOWNLOAD

Windows 10/11 (x64) & macOS (Apple Silicon)

Download: GitHub Releases (v1.0.0)

github.com/Azizsekerdil/smart-van-sales-management-system/releases

Two assets are published on GitHub Releases: a Windows zip and a macOS zip. The macOS package targets Apple Silicon (arm64) and is not notarized — on first launch, right-click → Open.

Smart Van Sales Management System

Every movement in the field recorded, every figure accounted for at day end

Install: `setup.ps1`

Start: `start.bat`

Guide: `docs/USER_GUIDE_EN.md`

This deck is generated from the product's own version

v1.0.0